🧠 **MACHINE LEARNING — COMPLETE EXAM TOPIC LIST**

🔹 **1. FOUNDATIONS**

- What is Machine Learning
- When to use / NOT use ML
- Generalization
- Overfitting vs underfitting
- Simplicity vs complexity

🔹 **2. DATA & PREPROCESSING**

- Feature engineering
- Feature selection
- One-hot encoding
- Handling missing data
- Scaling / normalization / standardization
- When scaling matters (distance-based models)
- Data preparation decisions (exam scenarios)

🔹 **3. SUPERVISED LEARNING (CORE SECTION)**

Models

- Linear regression
- Logistic regression
- Perceptron
- KNN

Model selection (VERY TESTED)

- Which models to use by default:
    - based on data type (tabular, high-dim, etc.)
    - interpretability requirements
    - dataset size

🔹 **4. LOSS FUNCTIONS**

- MSE
- MAE
- Cross-entropy

🔹 **5. OPTIMIZATION**

- Gradient descent
- Learning rate
- Convergence

🔹 **6. MODEL EVALUATION (VERY IMPORTANT)**

Confusion matrix

- TP, FP, FN, TN

Metrics

- Accuracy
- Precision
- Recall
- F1 score

Curves

- ROC curve
- Precision-Recall (PR) curve
- AUC

Critical understanding

- When to use ROC vs PR
- Effect of class imbalance
- Tradeoffs between precision/recall

🔹 **Metric Computation (EXAM SKILL)**

- Compute metrics from:
    - predictions
    - probabilities

- confusion matrix

◆ Decision Thresholding (VERY IMPORTANT)

- Selecting thresholds
- Operating point on ROC

Risk / Cost-based decisions

- Expected risk:

R=λ10PFP+λ01PFNR = \lambda_{10} P_{FP} + \lambda_{01} P_{FN}R=λ10PFP+λ01PFN

- Use:
    - class prevalence
    - cost matrix
- Choose optimal threshold

◆ 7. MODEL SELECTION & EXPERIMENT DESIGN

- Train / validation / test split
- k-fold cross validation
- LOOCV

Critical concepts

- When to use CV
- Performance evaluation vs comparison
- Experimental design

◆ Statistical Thinking (VERY TESTED)

- Variance in results
- Interpreting performance differences
- When is one model truly better?
- Performance under uncertainty

◆ 8. BIAS–VARIANCE TRADEOFF

- Bias
- Variance
- Noise
- Underfitting vs overfitting

◆ 9. DIMENSIONALITY REDUCTION

- PCA
- Eigenvectors / eigenvalues
- Projection
- t-SNE
- UMAP

◆ 10. REPRESENTATION LEARNING

- Autoencoders
- Bottleneck representations
- Masked autoencoders

◆ 11. DENSITY ESTIMATION

- p(x)p(x)p(x) vs p(y∣x)p(y|x)p(y∣x)
- Parametric methods
- Nonparametric methods
- Histograms
- KDE

◆ 12. GAUSSIAN MIXTURE MODELS (GMM)

- Mixture models
- Soft clustering
- EM algorithm
- Responsibilities
- Likelihood

◆ 13. CLUSTERING (VERY IMPORTANT)

Methods

- K-means
- Gaussian Mixture Models
- Hierarchical clustering
- DBSCAN
- Spectral clustering

Key comparisons (EXAM HEAVY)

- K-means vs GMM
- DBSCAN vs K-means
- Spectral vs K-means
- Hard vs soft clustering
- Density vs distance vs connectivity

Distance / similarity metrics

- Euclidean
- Manhattan
- Cosine

👉 Know:

- when to use each

Practical understanding

- Identify clustering method from output/behavior
- Match method to dataset shape

🔹 **14. REINFORCEMENT LEARNING (RL)**

Core components (VERY TESTED)

- Agent
- Environment
- State
- Action
- Reward
- Return
- Policy
- Value function
- Q-function

👉 Be able to map these to real scenarios

RL Models

- Markov Chains
- Markov Reward Processes (MRP)
- Markov Decision Processes (MDP)

Value Functions

- V(s)V(s)V(s)
- Q(s,a)Q(s,a)Q(s,a)

Bellman Equations

- Expectation
- Optimality

RL Algorithms

- Policy iteration
- Value iteration
- Monte Carlo
- Temporal Difference (TD)

TD Learning

- TD(0)
- Bootstrapping

Control Methods

- Q-learning (off-policy)
- SARSA (on-policy)

Exploration

- ε-greedy
- UCB

Deep RL

- DQN

🔥 15. KEY CONNECTIONS (VERY TESTABLE)

- Density estimation vs clustering
- K-means vs GMM
- DBSCAN vs K-means
- Spectral vs K-means
- ROC vs PR curves
- On-policy vs off-policy
- Monte Carlo vs TD learning